

Jake Brawer, Ph.D.

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RESEARCH OVERVIEW

Summary I build **human-robot collaborative systems** that learn through their interactions. My research draws upon recent innovations in machine learning as well as foundational techniques in artificial intelligence for designing systems that learn and utilize **abstract knowledge** to make social and physical interactions more **naturalistic** and **robust**.

EDUCATION

2023-Current **Postdoctoral Researcher** University of Colorado, Boulder
Advisors: Alessandro Roncone and Bradley Hayes

2016-2023 **Ph.D. in Computer Science** Yale University, Advisor: Brian Scassellati

2012-2016 **B.A. in Cognitive Science; Computer Science minor** Vassar College

SKILLS

Programming **Python**, **C-C++**, **L^AT_EX**, **Git**, **Jekyll**, **Emacs**, **Docker**, **Unity**

ML/AI/CV **Scikit-learn**, **Pytorch**, **numpy**, **pandas**, **matplotlib**, **NLTK**, **ROS**, **Gazebo**,
Tools **RViz**, **Gurobi**, **Prolog**, **PDDL**, **OpenCV**, **Open3D**

Robots and **Baxter**, **Sawyer**, **UR5**, **Kinect**, **RealSense**, **RGB-D** cameras
Sensors

System 10+ years of daily **Linux** experience

Soft Skills **Strong communicator**, **Loves collaboration**, **Fast learner**, **Leadership**

JOURNAL ARTICLES

Shivendra Agrawal, Ashutosh Naik, **Jake Brawer**, Alessandro Roncone, and Bradley Hayes (2025). “**ShelfAware: Semantic Particle Filter Localization with Low-Cost Sensors**”. In preparation. *A novel semantic Monte Carlo localization algorithm that requires just an RGB-D camera and a visual odometry camera*.

Meiying Qin, **Jake Brawer**, and Brian Scassellati (2022a). “**Robot Tool Use: A Survey**”. In: *Frontiers in Robotics and AI* 9, p. 369.

* Meiying Qin, * **Jake Brawer**, and Brian Scassellati (2021). “**Rapidly Learning Generalizable and Robot-Agnostic Tool-Use Skills for a Wide Range of Tasks**”. In: *Frontiers in Robotics and AI* 8, p. 380.

Jake Brawer, Aaron Hill, Ken Livingston, Eric Aaron, Joshua Bongard, and John H Long Jr (2017). “**Epigenetic Operators and the Evolution of Physically Embodied Robots**”. In: *Frontiers in Robotics and AI* 4, p. 1.

CONFERENCE PROCEEDINGS

- Jake Brawer**, Clare Lohrmann, Que Liu, Anthony Ries, Alessandro Roncone, and Bradley Hayes (2025). “**Effort-Aware Multi-Agent Task Allocation in Sequential Collaborative Scenarios**”. In preparation. *A decision support system can assist human users in regulating physical and cognitive effort during collaborative task sequences with autonomous agents*.
- Gyanig Kumar, Yashwant Gandham, Yi-Shiuan Tung, Anthony Ries, **Jake Brawer**, Alessandro Roncone, and Bradley Hayes (2025). “**Flexible Multimodal Shared Control for Assistive Robot Arm Manipulation**”. In preparation. *A multimodal shared autonomy system that combines eye gaze, joystick input, and semantic scene understanding to infer user intent and dynamically generate task-aligned control modes for assistive teleoperation*.
- Nikhil Hulle, Stéphane Aroca-Ouellette, Anthony J Ries, **Jake Brawer**, Katharina von der Wense, and Alessandro Roncone (2024). “**Eyes on the Game: Deciphering Implicit Human Signals to Infer Human Proficiency, Trust, and Intent**”. In: *33rd IEEE International Conference on Robot and Human Interactive Communication (RO-MAN 2024)*. Accepted for publication, proceedings pending.
- Jake Brawer**, Debasmita Ghose, Kate Candon, Meiying Qin, Alessandro Roncone, Marynel Vazquez, and Brian Scassellati (2023). “**Interactive Policy Shaping for Human-Robot Collaboration with Transparent Matrix Overlays**”. In: *Proceedings of the 2023 ACM/IEEE International Conference on Human-Robot Interaction*. IEEE.
- Meiying Qin, **Jake Brawer**, and Brian Scassellati (2022b). “**Task-Oriented Robot-to-Human Handovers in Collaborative Tool-Use Tasks**”. In: *2022 31st IEEE International Conference on Robot & Human Interactive Communication (RO-MAN)*. IEEE.
- Jake Brawer**, Meiying Qin, and Brian Scassellati (2020). “**A causal approach to tool affordance learning**”. In: *2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, pp. 8394–8399.
- Brian Scassellati, **Jake Brawer**, Katherine Tsui, Setareh Nasihati Gilani, Melissa Malzkuhn, Barbara Manini, Adam Stone, Geo Kartheiser, Arcangelo Merla, Ari Shapiro, et al. (2018). “**Teaching Language to Deaf Infants with a Robot and a Virtual Human**”. In: *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. ACM, p. 553.
- Zong Xuan Tan, **Jake Brawer**, and Brian Scassellati (2018). “**That’s Mine! Learning Ownership Relations and Norms for Robots**”. In: *Thirty-second AAAI conference on artificial intelligence*.

WORKSHOP PROCEEDINGS

- Ava Abderezaei, Chi-Hui Lin, Joseph Miceli, Naren Sivagnanadasan, Stéphane Aroca-Ouellette, **Jake Brawer**, and Alessandro Roncone (2025). “**Towards Zero-Shot Coordination between Teams of Agents: The N-XPlay Framework**”. In: *Workshop on Multi-Robot Systems. Robotics: Science and Systems (RSS)*. Workshop paper; proceedings forthcoming.
- Jake Brawer**, Kayleigh Bishop, Bradley Hayes, and Alessandro Roncone (2023). “**Towards A Natural Language Interface for Flexible Multi-Agent Task Assignment**”. In: *Proceedings of the AAAI Symposium Series*. Vol. 2. 1, pp. 167–171.

AWARDS AND FELLOWSHIPS

- July 2023, 2024, 2025 **Army Educational Outreach Program (AEOP) Fellowship**, University of Colorado Boulder — awarded in 2023, 2024, and 2025 to support postdoctoral research.
- March 2023 **Best paper (technical track):** Our paper “Interactive Policy Shaping for Human-Robot Collaboration” won the best paper award in the technical track at the 2023 International Conference on Human-Robot Interaction (HRI 2023).

WORKSHOPS ORGANIZED

- March 2023 **Human-Machine Teaming Paradigm Workshop**
Jake Brawer, Bradley Hayes, Alessandro Roncone
- March 2018 **Bridging the Gap: An NSF Workshop on Conversational Agents and Human-Robot Interaction**
Justine Cassell, Brian Scassellati, Jake Brawer, Michael Madaio
NSF Cyber-Human Systems (CHS), Robust Intelligence, National Robotics Initiative. Award #1829237

TEACHING EXPERIENCE

- Spring 2024 **CSCI 4302/5302 Advanced Robotics**
Sole Lecturer. Topics covered included motion planning, SLAM, optimization, and reinforcement learning, culminating in a final project.
- Spring 2018/2019 **Intelligent Robotics**
Teaching Assistant
- Fall 2017 **Object Oriented Programming**
Teaching Assistant
- Fall 2014/2015 **Perception and Action**
Teaching Assistant

SERVICE

- Leadership Area Chair, International Conference on Human-Agent Interaction (**HAI**; 2024)
- Conference reviews Robotics: Science and Systems (**RSS**; 2025 - **Outstanding Reviewer Award (2025)**)
International Conference on Humanoid Robots (**Humanoids**; 2018)
International Conference on Intelligent Robots and Systems (**IROS**; 2020)
International Conference on Human-Robot Interaction (**HRI**; 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2025 - **special mention for Outstanding Review (2023)**)
International Conference on Robotics and Automation (**ICRA**; 2019)

Journal	International Journal of Robotics Research (IJRR ; 2024)
reviews	IEEE Transaction on Robotics (T-RO ; 2024)
	ACM Transactions on Human–Robot Interaction (THRI ; 2019, 2020, 2024, 2025)
Students	Shivendra Agrawal (2024-2025) Ava Abderezaei (2024-2025) Gyaning Kumar (2025)
supervised	Yi-Shiuan Huang (2025) Chi-Hui Lin (2024-2025) Aaquib Tabrez (2023-2024) Kaleb Bishop (2017-2019, 2023) John Dallard (2021-2022) Sarah Widder (2017-2019) Kevin Choi (2018) Acshi Haggemiller (2016-2017) Tan Zong Xuan (2017-2018)